

KINGSFORD, MI, USA

WMO: 727437

Lat: 45.818N

Lon: 88.114W

Elev: 1122

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 94893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-6.9	-18.8	2.0	-9.7	-14.7	2.6	-5.5	24.6	17.8	21.5	16.5	3.7	280	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	88.0	70.7	84.5	68.4	81.9	67.0	73.6	82.8	71.4	80.4	69.5	77.9	10.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.5	117.1	78.4	68.5	109.0	76.3	66.3	100.8	74.4	37.9	82.7	35.9	80.2	34.3	78.6	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.6	16.6	-19.3	93.0	4.6	2.9	-22.5	95.1	-25.2	96.8	-27.7	98.4	-31.1	100.6		
			-19.6	76.8	4.4	2.3	-22.8	78.5	-25.3	79.8	-27.8	81.1	-31.0	82.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	15.3	17.6	27.9	40.6	53.3	63.1	67.8	65.4	58.1	45.2	32.5	20.8		
	DBStd	20.64	11.32	11.07	11.53	9.07	8.78	7.31	6.33	6.02	8.10	8.67	9.22	10.75		
	HDD50	4704	1076	907	692	304	64	3	0	0	19	204	528	907		
	HDD65	8559	1541	1327	1152	733	376	121	45	68	234	616	974	1372		
	CDD50	1940	0	0	6	21	167	395	551	478	264	55	3	0		
	CDD65	319	0	0	0	0	15	62	130	81	29	2	0	0		
	CDH74	3706	0	0	5	22	269	770	1470	837	308	25	0	0		
	CDH80	1050	0	0	0	4	73	224	474	203	69	3	0	0		
	WSAvg	6.8	7.3	7.3	7.2	7.6	7.4	6.3	5.9	5.5	6.2	6.8	7.1	6.9		
	PrecAvg	28.0	1.2	0.9	1.5	2.2	3.1	3.3	3.2	3.3	3.4	2.6	1.8	1.4		
Precipitation	PrecMax	37.0	4.0	2.5	4.1	4.4	6.7	6.5	11.3	6.9	6.1	6.3	5.4	3.5		
	PrecMin	18.7	0.1	0.0	0.2	0.2	0.4	1.0	0.9	0.2	0.7	0.4	0.3	0.2		
	PrecStd	4.5	0.8	0.7	1.0	1.0	1.5	1.3	1.7	1.3	1.5	1.4	1.1	0.7		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	49.6	69.5	77.1	87.1	90.9	92.6	89.8	87.2	77.9	63.9	48.3		
		MCWB	36.5	40.7	57.3	58.3	66.0	72.7	73.3	72.0	70.2	63.1	52.8	42.3		
	2%	DB	37.4	41.9	56.8	69.8	81.3	85.9	88.3	84.7	81.4	71.1	55.2	41.1		
		MCWB	34.0	36.7	47.6	53.7	62.6	68.6	71.2	69.1	67.1	59.0	48.7	37.2		
	5%	DB	34.4	36.8	50.0	63.7	75.8	82.1	85.1	81.8	77.5	65.4	50.4	37.1		
		MCWB	32.1	33.1	41.7	49.0	59.6	67.0	69.3	67.3	64.7	57.8	44.7	34.6		
	10%	DB	31.9	33.9	44.7	58.5	70.9	78.8	81.9	79.0	73.1	60.2	46.2	34.6		
		MCWB	30.0	31.2	38.3	45.9	57.1	64.0	67.6	65.5	63.0	53.4	42.0	32.7		
	0.4%	WB	37.3	42.4	57.7	60.7	69.8	75.1	77.4	75.3	72.7	67.9	55.0	43.7		
		MCDB	40.7	48.5	64.5	71.9	80.4	84.9	87.2	83.9	81.1	72.8	60.7	45.3		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	34.5	37.0	49.3	55.3	66.1	71.6	74.1	72.4	69.7	61.8	50.3	37.9		
		MCDB	37.0	41.0	55.6	66.6	76.2	81.6	83.6	80.6	77.6	68.5	55.1	40.3		
	5%	WB	32.4	33.5	43.0	50.6	62.7	68.9	71.9	70.3	67.5	57.9	45.7	35.1		
		MCDB	34.4	36.3	48.8	60.1	71.3	77.5	80.8	78.1	73.8	63.6	49.0	36.9		
	10%	WB	29.9	30.9	38.5	47.1	59.2	66.4	69.7	68.2	65.0	53.9	42.1	32.7		
		MCDB	31.9	33.6	43.8	57.0	68.7	74.6	78.9	75.6	70.9	59.1	46.2	34.7		
Mean Daily Temperature Range	5% DB	MDBR	16.0	18.9	20.3	23.3	25.0	23.8	24.3	23.2	22.9	19.3	15.0	14.3		
		MCDBR	16.5	20.5	26.7	34.9	32.1	27.6	27.0	26.3	27.7	27.3	22.9	16.7		
		MCWBR	14.1	15.8	17.6	20.3	16.2	13.3	12.6	12.6	14.5	17.0	16.3	13.6		
	5% WB	MCDBR	15.8	18.1	24.2	30.9	26.6	23.9	23.9	21.7	21.9	23.8	20.2	14.8		
		MCWBR	13.9	14.3	16.8	19.6	14.8	13.3	12.6	11.5	13.2	16.7	15.6	12.6		
Clear-Sky Solar Irradiance	taub		0.242	0.275	0.313	0.338	0.373	0.371	0.371	0.372	0.350	0.314	0.286	0.254		
	taud		2.425	2.317	2.360	2.405	2.355	2.383	2.403	2.393	2.462	2.557	2.540	2.462		
	Ebn at Noon		282	288	291	292	284	284	282	278	276	273	263	265		
	Edh at Noon		22	30	33	35	38	37	36	35	30	23	20	19		
All-Sky Solar Radiation	RadAvg		437	759	1160	1449	1731	1868	1926	1627	1247	760	436	319		
	RadStd		41	85	110	142	141	148	94	106	83	82	47	33		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page